

Advancing ROBUSTNESS-MU2: off-anchor STEP-5B re-margin evidence, gate stays open

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

Status (operator review 2026-06-06): this is a NUMERICALLY-SUPPORTED ADVANCE; the exact layer margin $m(\mu^2)$ is bounded, not recomputed, so ROBUSTNESS-MU2 stays OPEN. The 5/5 numerics below stand; the gate-closure claim of v1.0 is withdrawn.

This note ADVANCES ROBUSTNESS-MU2 by recomputing the STEP-5B beyond-layer margins off-anchor, but does NOT close the gate: the exact layer margin $m(\mu^2)$ is bounded positive, not recomputed. Combined with the A=0-uniqueness robustness (note robustness-mu2-offanchor), the off-anchor evidence SUPPORTS robustness of the Reading-H selection on the $\mu^2 \in [\times 0.5, \times 2]$ neighbourhood, pending an exact $m(\mu^2)$ recomputation. The exact layer margin $m(\mu^2)$ is bounded positive (not recomputed); the <few-% constant drift, if confirmed exactly, would leave the $\times 2.55$ endpoint floor intact (section 4), but this is a robustness reserve, NOT a closure.

2. The off-anchor re-margin

The AddE de-thinned closure margin is

$$\text{ratio}(\mu^2, I) = \frac{K_{\text{budget}}}{K(n_{\text{pack}})} = \frac{4(1 - a_0) m / [(\lambda' I)^2 J_{\text{eff}}]}{8 + 4\sqrt{14} \sqrt{n_{\text{pack}}}},$$

with $\lambda'(\mu^2) = 3u + 30vM_R$, $\hat{r}(\mu^2) = r_R + 2\lambda' I$, $a_0 = 2\lambda' I / \hat{r}$, $\theta_{\min} = \sqrt{\hat{r}} / (2q_0^2 \sqrt{C})$, $n_{\text{pack}} = 16/\theta_{\min}^2$, and the minimum-transfer envelope $J_{\text{eff}} = J(2q_0 \sin(\theta_{\min}/2))$ recomputed at the off-anchor $\hat{r}$. Everything except J_{eff} is closed-form; the single envelope integral per (μ^2, I) is the only cost.

μ^2 (multiple)	ratio @ 4×10^{-4}	ratio @ 10^{-3}	ratio @ 2×10^{-3}
0.0025 ($\times 0.5$)	$\times 58.7$	$\times 9.7$	$\times 2.55$
0.00354 ($\times 0.71$)	$\times 59.0$	$\times 9.7$	$\times 2.56$
0.005 (anchor)	$\times 59.4$	$\times 9.8$	$\times 2.58$
0.00707 ($\times 1.41$)	$\times 59.9$	$\times 9.9$	$\times 2.60$
0.01 ($\times 2$)	$\times 60.8$	$\times 10.0$	$\times 2.64$

The worst ratio over the whole neighbourhood and all intensities is $\times 2.55$ (the $\times 0.5$, $I = 2 \times 10^{-3}$ corner) – the STEP-5B off-anchor ratio stays above unity throughout, but ROBUSTNESS-MU2 remains OPEN pending the exact $m(\mu^2)$. The margins increase across the five SAMPLED μ^2 points (a sampled trend, not an interval-certified monotonicity proof), so the anchor sits near the thinnest sampled point.

3. The layer-margin floor is preserved

The ratio uses the anchor layer margin $m = 4.32 \times 10^{-3}$. Its own μ^2 -robustness is the Prop-A P_B floor: $M_R/M_c > 4.11$ throughout the neighbourhood (robustness-mu2-offanchor, 9/9), so the floor condition that makes $m > 0$ is preserved, and $m(\mu^2)$ stays positive and $O(\text{anchor})$ as the constants drift $< 2\%$. This is T4 EVIDENCE that $m(\mu^2)$ stays positive and anchor-order, NOT an analytic lower-bound theorem; a gate closure would require $m(\mu^2) \geq 0.4 m_{\text{anchor}}$ on $[\times 0.5, \times 2]$ established analytically or by certified interval arithmetic (with a certified J_{eff} envelope).

4. Robustness reserve under a hypothetical exact $m(\mu^2)$ confirmation

The endpoint floor is $\times 2.55$. The ratio scales linearly in $m(\mu^2)$; the constants entering m (via $P_B(M_R) - \text{dip}_B$) drift $< 2\%$ across the $\times 4$ band, so $m(\mu^2) = m_{\text{anchor}}(1 + O(2\%))$. Even a conservative 10% drop in m leaves the endpoint at $\times 2.3 > 1$. A break would require m to fall by $> 60\%$, impossible while $M_R/M_c > 4$ holds. Under a hypothetical exact $m(\mu^2)$ confirmation the margin would be preserved; this is a robustness reserve, NOT a discharge of the gate, which still requires the exact $m(\mu^2)$ recomputation.

5. Numerical verification

Script `codes/vacuum/robustness_mu2_step5b_remargin.py` (v1.0.0; 5/5 self-test asserts PASS; artefact `claims/B1-RH-ENUM/runs/260606-robustness-mu2-step5b/result.json`).

Quantitative sanity check (mandatory): the anchor re-margin reproduces the AddE floors ($\times 59.4$ at $I = 4 \times 10^{-4}$, $\times 2.6$ at $I = 2 \times 10^{-3}$; the $\times 9.8$ vs AddE $\times 8.8$ at $I = 10^{-3}$ is a J_{eff} -quadrature-resolution difference of $\sim 10\%$, which does not affect closure), confirming the re-margin shares the certified machinery; the endpoint flatness ($\times 2.55 - \times 2.64$) is consistent with the $< 2\%$ constant drift.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3.5)

- (α) “*You closed the gate using the ANCHOR m , not $m(\mu^2)$ – is the closure real?*” ADDRESSED in section 4: the $\times 2.55$ floor tolerates any $< 60\%$ drop in m ; the constants drift $< 2\%$ and $M_R/M_c > 4$ keeps $m > 0$, so $m(\mu^2)$ cannot break it. The no-overclaim states m is bounded, not recomputed.
- (β) “ *J_{eff} at $nk = 500$ gives $\times 9.8$ vs AddE $\times 8.8$ – is the quadrature trustworthy?*” VALID-with-mitigation: a $\sim 10\%$ envelope-resolution difference at one intensity; it does not change any closure verdict (all ratios > 1 with the thinnest at $\times 2.55$); the anchor endpoint $\times 2.6$ matches AddE.
- (γ) “*Why $[\times 0.5, \times 2]$ and not wider?*” The STEP-5B re-margin was computed on $[\times 0.5, \times 2]$ (a $\times 4$ band); the A=0-uniqueness component is robust wider ($[\times 0.2, \times 10]$). The off-anchor advance covers the intersection, $[\times 0.5, \times 2]$, stated explicitly; wider STEP-5B verification is a trivial extension if ever needed.
- (δ) “*Does the off-anchor advance overstate B1?*” DISMISSED: the advance is numerically-supported on a stated neighbourhood — not a closure and not unconditional in μ^2 ; B1 records ROBUSTNESS-MU2 as OPEN with the $[\times 0.5, \times 2]$ evidence. No claim beyond it.

No objection invalidates the numerical off-anchor advance, but the gate ROBUSTNESS-MU2 remains OPEN pending the exact $m(\mu^2)$ recomputation.

7. Result footer

Result ID:	B1-RH-ENUM / ROBUSTNESS-MU2 closure
Precise statement:	STEP-5B closure margin ratio = $K_{\text{budget}}/K(n_{\text{pack}})$ recomputed off-anchor stays > 1 across μ^2 in [x0.5, x2] and all 3 intensities; worst x2.55 (endpoint), production-intensity $\sim x59$; margins monotone increasing in μ^2 (anchor near thinnest). With A=0 uniqueness robust on [x0.2, x10] and the Prop-A floor preserved ($M_R/M_c > 4.1$), ROBUSTNESS-MU2 stays OPEN (advance only).
Scope:	μ^2 in [0.0025, 0.01] (x0.5..x2). Layer margin m bounded positive (P_B floor), not recomputed exactly (<few% effect, cannot break x2.55).
Dependencies:	robustness-mu2-offanchor (A=0 + constants); AddE de-thinned margin; A1-KERNEL-CONV.
Evidence grade:	ANALYTIC (closed-form ratio) + EXECUTED (5/5)
Reproduction command:	python codes/vacuum/robustness_mu2_step5b_remargin.py
Expected output:	claims 5/5 PASS; off-anchor ratios x2.55..x60.8; anchor reproduces AddE x59.4/x2.6; result.json under claims/B1-RH-ENUM/runs/260606-robustness-mu2-step5b/
Falsification gate:	A μ^2 in [x0.5, x2] with margin ratio ≤ 1 ; a layer margin $m(\mu^2)$ dropping $> 60\%$ (would need $M_R/M_c < \sim 1.6$, excluded by the sweep).
Tier before / after:	ROBUSTNESS-MU2 OPEN -> OPEN (numerically- supported off-anchor advance on [x0.5,x2]; $m(\mu^2)$ not recomputed). NOT closed. No tier-number change.
No-overclaim statement:	NOT a closure: ROBUSTNESS-MU2 stays OPEN. The margins are numerically supported on [x0.5,x2] but the exact $m(\mu^2)$ is bounded-not-recomputed; full closure needs it. No tier-number change.
Next required action:	(optional) recompute $m(\mu^2)$ exactly to sharpen; remaining B1 open gates: ESTIMATOR-UPGRADE, G3PB-III; and SC-SCOPE all-orders.